Linear inequalities in one or two variables

Algebra · Math · 77 questions

1 0fd50277-112f-40b1-b403-c45b8c0bef89 · Easy

Leo goes to a packing store to buy containers and tape. Leo has $\$ 15$. Each container costs $\$ 1.87$ and each roll of tape costs $\$ 2.40$. Which inequality represents the relationship between the number of containers, c, and the number of rolls of tape, t, Leo can buy?

A. $1.87c + 2.40t \leq 15$

B. $1.87c + 2.40t \geq 15$

C. $2.40c + 1.87t \leq 15$

D. $2.40c + 1.87t \geq 15$

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2 09101-DC · Easy

Valentina bought two containers of beads. In the first container 30% of the beads are red, and in the second container 70% of the beads are red. Together, the containers have at least 400 red beads. Which inequality shows this relationship, where x is the total number of beads in the first container and y is the total number of beads in the second container?

A. $0.3x + 0.7y \geq 400$

B. $0.7x + 0.3y \leq 400$

C. $\frac{x}{3} + \frac{y}{7} \leq 400$

D. $30x + 70y \geq 400$

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3 43bbec1d-b28f-43e6-89b0-1582a7e5c4a3 · Easy

Julissa needs at least 100 hours of flight time to get her private pilot certification. If Julissa already has 86 hours of flight time, what is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification?

A. 14

B. 76

C. 86

D. 186

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4 d6f43e04-786e-41b9-a88e-42db5ecb8f8b · Easy

A cleaning service that cleans both offices and homes can clean at most 14 places per day. Which inequality represents this situation, where f is the number of offices and h is the number of homes?

A. $f + h \leq 14$

B. $f + h \geq 14$

C. $f - h \leq 14$

D. $f - h \geq 14$

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5 1d4cae8e-913c-4a09-b279-259bf349f1f5 · Easy

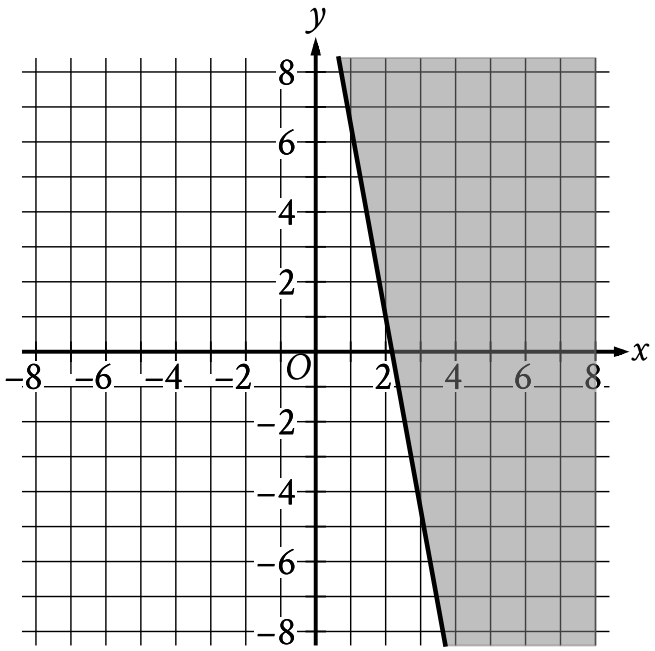
For a snowstorm in a certain town, the minimum rate of snowfall recorded was 0.6 inches per hour, and the maximum rate of snowfall recorded was 1.8 inches per hour. Which inequality is true for all values of s, where s represents a rate of snowfall, in inches per hour, recorded for this snowstorm?

A. $s \geq 2.4$

B. $s \geq 1.8$

C. $0 \leq s \leq 0.6$

D. $0.6 \leq s \leq 1.8$



- The boundary of the inequality is a solid line.
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (1.0 comma 6.5)
    - (3.0 comma negative 4.5)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents solutions to an inequality. Which ordered pair x, y is a solution to this inequality?

A. $0, -4$

B. $0, 4$

C. $-4, 0$

D. $4, 0$

7 d9aed3a1-8247-4847-89fd-1c615ddd8b29 · Easy

Ty set a goal to walk at least 24 kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of 4 kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

A. 4

B. 6

C. 20

D. 24

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8 784ebc69-0579-465e-9969-ad19889a3bd4 · Easy

For a party, 50 dinner rolls are needed. Dinner rolls are sold in packages of 12. What is the minimum number of packages that should be bought for the party?

Student-produced response: _____

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9 7d42453b-bbe3-4173-bd1f-6c5cbeaac95f · Easy

A tow truck travels at a speed of at least 25 miles per hour but no more than 45 miles per hour for a section of a road. Which inequality represents this situation, where x is the speed of the tow truck, in miles per hour, on this section of the road?

A. $x \geq 25$

B. $x \geq 45$

C. $25 \leq x \leq 45$

D. $x \leq 70$

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10 025731-DC · Easy

A clothing store is having a sale on shirts and pants. During the sale, the cost of each shirt is $15 and the cost of each pair of pants is $25. Geoff can spend at most $120 at the store. If Geoff buys s shirts and p pairs of pants, which of the following must be true?

A. $15s + 25p \leq 120$

B. $15s + 25p \geq 120$

C. $25s + 15p \leq 120$

D. $25s + 15p \geq 120$

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11 dcb9ea1e-c069-4c4c-92f7-b7633bfd6904 · Easy

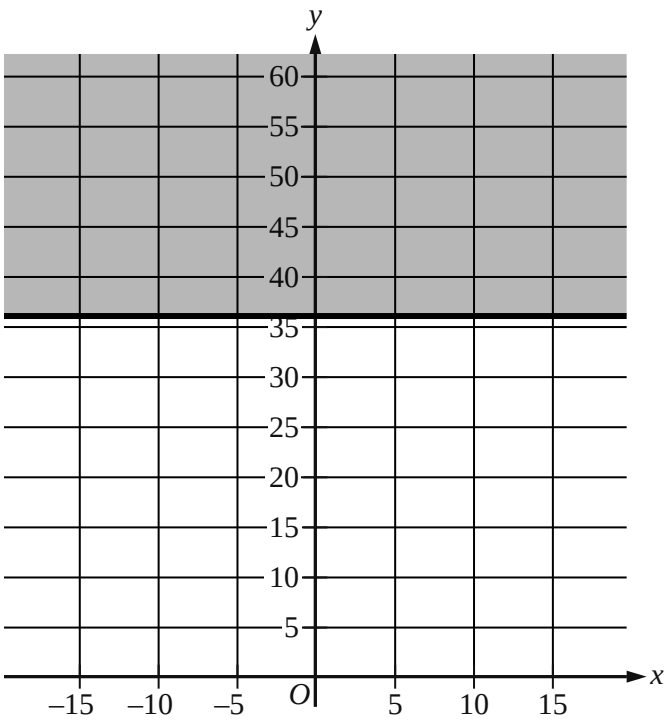
During spring migration, a dragonfly traveled a minimum of $1{,}510$ miles and a maximum of $4{,}130$ miles between stopover locations. Which inequality represents this situation, where d is a possible distance, in miles, this dragonfly traveled between stopover locations during spring migration?

A. $d \leq 1{,}510$

B. $1{,}510 \leq d \leq 4{,}130$

C. $d \geq 4{,}130$

D. $4{,}130 \leq d \leq 5{,}640$



- The boundary of the inequality is a solid line.
  - The line is horizontal.
  - The line passes through the following points:
    - (negative 15 comma 36)
    - (0 comma 36)
    - (15 comma 36)
- The area above the boundary is shaded.

The shaded region shown in the graph represents all the solutions to which inequality?

A. $x \leq 36$

B. $x \geq 36$

C. $y \leq 36$

D. $y \geq 36$

13 182f4a98-459c-4659-9484-08aa8d678af0 · Easy

During a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

A. $s \leq 20$

B. $s \leq 150$

C. $s \leq 170$

D. $150 \leq s \leq 170$

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14 3f8fee7b-88df-4e61-b457-3a88934cb9c7 · Easy

For a 3-week period in a town in Illinois, the lowest recorded temperature was 31 degrees Fahrenheit $°\mathrm{F}$ and the highest recorded temperature was $67°\mathrm{F}$. Which inequality is true for any recorded temperature t, in $°\mathrm{F}$, in this town for this 3-week period?

A. $t \geq 98$

B. $t \geq 67$

C. $31 \leq t \leq 67$

D. $t \leq 31$

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15 782fda69-e7ad-4084-b602-feec64776b52 · Easy

For a literature class, a student's essay needs to have no more than $5{,}000$ words. If a student's essay already has $3{,}026$ words, what is the maximum number of additional words the student can include in the essay?

A. $1{,}974$

B. $3{,}026$

C. $5{,}000$

D. $8{,}026$

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16 04176-DC · Easy

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

1. $(1, 1)$
2. $(2, 5)$
3. $(3, 2)$

A. I only

B. II only

C. I and II only

D. I and III only

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17 c6ae5b6e-3980-49ff-bd21-fd706a9011aa · Easy

$$2l + 2w \leq 27$$

A rectangle has length l and width w. The inequality gives the possible values of l and w for which the perimeter of this rectangle is less than or equal to 27. Which statement is the best interpretation of $l, w = 8, 3$ in this context?

A. If the rectangle has length 3 and width 8, its perimeter is less than or equal to 27.

B. If the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.

C. If the rectangle has length 3 and width 8, its perimeter is greater than or equal to 27.

D. If the rectangle has length 8 and width 3, its perimeter is greater than or equal to 27.

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18 06905-DC · Easy

A bakery sells trays of cookies. Each tray contains at least 50 cookies but no more than 60. Which of the following could be the total number of cookies on 4 trays of cookies?

A. 165

B. 205

C. 245

D. 285

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19 025455-DC · Easy

Normal body temperature for an adult is between $97.8°F$ and $99°F$, inclusive. If Kevin, an adult male, has a body temperature that is considered to be normal, which of the following could be his body temperature?

A. $96.7°F$

B. $97.6°F$

C. $97.9°F$

D. $99.7°F$

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20 5033c9a1-3f55-43c6-abad-0ff80dc97762 · Easy

Monarch butterflies can fly only with a body temperature of at least 55.0 degrees Fahrenheit $°F$. If a monarch butterfly's body temperature is $51.3°F$, what is the minimum increase needed in its body temperature, in $°F$, so that it can fly?

A. 1.3

B. 3.7

C. 5.0

D. 6.3

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21 05401-DC · Easy

Maria plans to rent a boat. The boat rental costs $60 per hour, and she will also have to pay for a water safety course that costs $10. Maria wants to spend no more than $280 for the rental and the course. If the boat rental is available only for a whole number of hours, what is the maximum number of hours for which Maria can rent the boat?

Student-produced response: _____

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22 be93eda4-1b01-4cb3-9e00-0859362e1713 · Easy

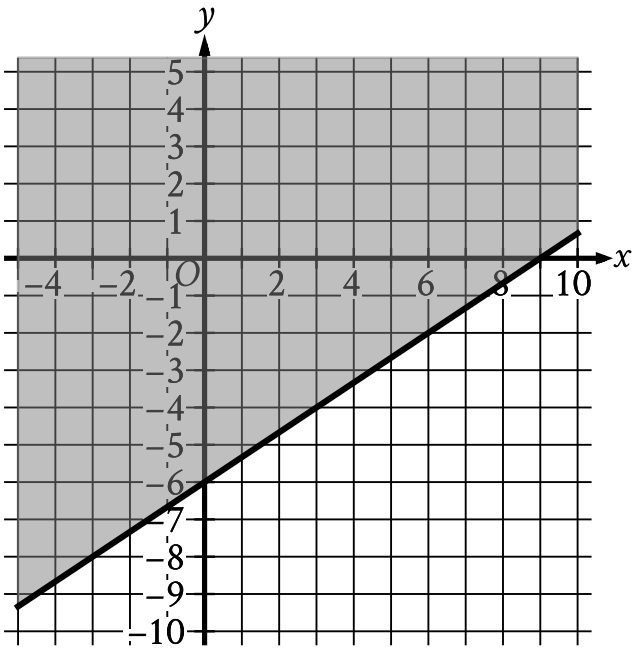
The total cost, in dollars, to rent a surfboard consists of a $\$ 25$ service fee and a $\$ 10$ per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of $\$ 75$ to rent the surfboard. Which inequality represents this situation?

A. $10t \leq 75$

B. $10 + 25t \leq 75$

C. $25t \leq 75$

D. $25 + 10t \leq 75$



- The boundary of the inequality is a solid line.
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (0 comma negative 6)
    - (9 comma 0)
- The area above and to the left of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y \geq \frac{2}{3}x - 6$

B. $y \geq \frac{2}{3}x + 6$

C. $y \geq \frac{2}{3}x - 9$

D. $y \geq \frac{2}{3}x + 9$

24 deb03694-cf78-40ff-a84f-f782eba08ba4 · Easy

The point $(8, 2)$ in the xy-plane is a solution to which of the following systems of inequalities?

A. $x > 0$ $y > 0$

B. $x > 0$ $y < 0$

C. $x < 0$ $y > 0$

D. $x < 0$ $y < 0$

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25 09378-DC · Easy

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G, on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

A. $90-(85+78+98) \leq 4G$

B. $4G+85+78+98 \geq 360$

C. $\frac{(G+85+78+98)}{4} \geq 90$

D. $\frac{(85+78+98)}{4} \geq 90-4G$

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26 a0522055-0d67-4148-a6b2-7e78be641a3b · Easy

A geologist needs to collect at least 67 samples of lava from a volcano. If the geologist has already collected 63 samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

A. 130

B. 63

C. 4

D. 0

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27 14928-DC · Easy

An elementary school teacher is ordering x workbooks and y sets of flash cards for a math class. The teacher must order at least 20 items, but the total cost of the order must not be over $80. If the workbooks cost $3 each and the flash cards cost $4 per set, which of the following systems of inequalities models this situation?

A. $x + y \geq 20$
$3x + 4y \leq 80$

B. $x + y \geq 20$
$3x + 4y \geq 80$

C. $3x + 4y \leq 20$
$x + y \geq 80$

D. $x + y \leq 20$
$3x + 4y \geq 80$

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28 816cb6b4-6796-4c47-ae29-2ccb52e5c8ab · Easy

On a car trip, Rhett and Jessica each drove for part of the trip, and the total distance they drove was under 220 miles. Rhett drove at an average speed of 35 miles per hour mph, and Jessica drove at an average speed of 40 mph. Which of the following inequalities represents this situation, where r is the number of hours Rhett drove and j is the number of hours Jessica drove?

A. $35r + 40j > 220$

B. $35r + 40j < 220$

C. $40r + 35j > 220$

D. $40r + 35j < 220$

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29 40468143-fc48-4d07-9364-198a459e602f · Easy

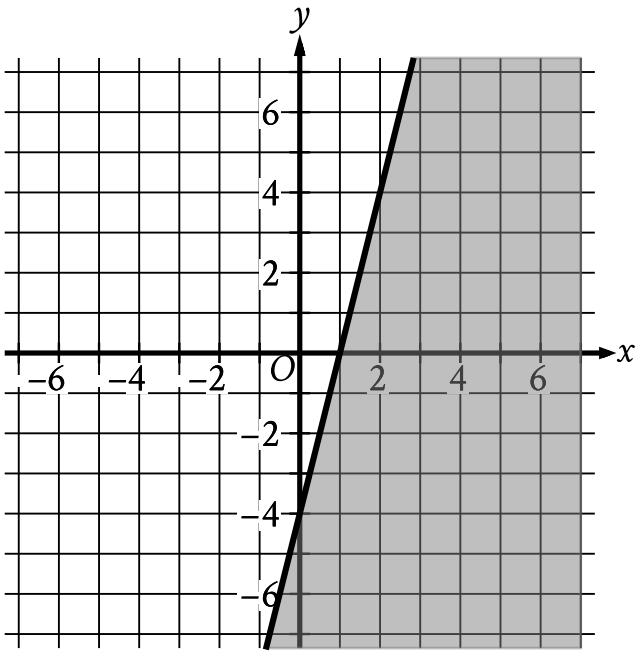
A geologist estimates that the volume of a slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet. The geologist also estimates that the slab of granite weighs 165 pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the slab of granite?

A. $165 - 15.7 < x < 165 - 12.7$

B. $165 + 12.7 < x < 165 + 15.7$

C. $16512.7 < x < 16515.7$

D. $\frac{165}{15.7} < x < \frac{165}{12.7}$



- The boundary of the inequality is a solid line.
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma negative 4)
    - (1 comma 0)
    - (2 comma 4)
- The area below and to the right of the boundary is shaded.

The shaded region shown represents the solutions to an inequality. Which ordered pair x, y is a solution to this inequality?

A. $-5, -6$

B. $-2, 5$

C. $1, 4$

D. $6, -2$

31 20d7e793-fc4b-46cb-b6a6-4388c4317d68 · Medium

A particular botanist classifies a species of plant as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h, in centimeters, for a specific plant species when fully grown. Which inequality represents the possible heights h, in centimeters, for a tall plant species?

A. $106 < h < 158$

B. $80 < h < 100$

C. $42 < h < 87$

D. $17 < h < 85$

$$y > 7x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	13
5	27
8	48

B.

x	y
3	17
5	31
8	52

C.

x	y
3	21
5	27
8	52

D.

x	y
3	21
5	35

8	56

33 bdad3961-fd1e-41f1-af28-fb816aa8a0b1 · Medium

A city employee will plant two types of bushes, azaleas and boxwoods, in a park. There will be no more than 164 total bushes planted, and the number of azaleas planted will be at most three times the number of boxwoods planted. Which of the following systems of inequalities best represents this situation, where a is the number of azaleas that will be planted, and b is the number of boxwoods that will be planted?

A. $a + b \geq 164$ $3a \geq b$

B. $a + b \geq 164$ $a \leq 3b$

C. $a + b \leq 164$ $3a \geq b$

D. $a + b \leq 164$ $a \leq 3b$

34 4d535a61-d332-4981-961c-a1bf2488438f · Medium

The minimum value of x is 12 less than 6 times another number n. Which inequality shows the possible values of x?

A. $x \leq 6n - 12$

B. $x \geq 6n - 12$

C. $x \leq 12 - 6n$

D. $x \geq 12 - 6n$

35 271983e5-3bd9-4ac2-96fd-0b6b9aeca483 · Medium

In the xy-plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

A. The x-axis

B. The region where $x > 0$ and $y > 0$

C. The region where $x < 0$ and $y < 0$

D. The region where $x < 0$ and $y > 0$

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36 f1ec7cf0-eab8-4ce6-b73f-e779a85c2edc · Medium

The length of a rectangle is 50 inches and the width is x inches. The perimeter is at most 210 inches. Which inequality represents this situation?

A. $2x + 100 \leq 210$

B. $2x + 100 \geq 210$

C. $2x + 50 \leq 210$

D. $2x + 50 \geq 210$

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37 022207-DC · Medium

In North America, the standard width of a parking space is at least 7.5 feet and no more than 9.0 feet. A restaurant owner recently resurfaced the restaurant's parking lot and wants to determine the number of parking spaces, n, in the parking lot that could be placed perpendicular to a curb that is 135 feet long, based on the standard width of a parking space. Which of the following describes all the possible values of n ?

A. $18 \leq n \leq 135$

B. $7.5 \leq n \leq 9$

C. $15 \leq n \leq 135$

D. $15 \leq n \leq 18$

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38 ff06a0a2-65a5-41cc-a2c6-f909deb61b07 · Medium

An event planner is planning a party. It costs the event planner a onetime fee of $\$ 35$ to rent the venue and $\$ 10.25$ per attendee. The event planner has a budget of $\$ 200$. What is the greatest number of attendees possible without exceeding the budget?

Student-produced response: _____

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39 028905-DC · Medium

A certain elephant weighs 200 pounds at birth and gains more than 2 but less than 3 pounds per day during its first year. Which of the following inequalities represents all possible weights w, in pounds, for the elephant 365 days after birth?

A. $400 < w < 600$

B. $565 < w < 930$

C. $730 < w < 1{,}095$

D. $930 < w < 1{,}295$

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40 10570-DC · Medium

The average annual energy cost for a certain home is $4,334. The homeowner plans to spend $25,000 to install a geothermal heating system. The homeowner estimates that the average annual energy cost will then be $2,712. Which of the following inequalities can be solved to find t, the number of years after installation at which the total amount of energy cost savings will exceed the installation cost?

A. $25{,}000 > (4{,}334 - 2{,}712)t$

B. $25{,}000 < (4{,}334 - 2{,}712)t$

C. $25{,}000 - 4{,}334 > 2{,}712t$

D. $25{,}000 > \frac{4{,}332}{2{,}712}t$

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41 032729-DC · Medium

$y \leq x$

$y \leq -x$

Which of the following ordered pairs (x,y) is a solution to the system of inequalities above?

A. $(1,0)$

B. $(-1,0)$

C. $(0,1)$

D. $(0,-1)$

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42 10589-DC · Medium

$y \leq 3x+1$
$x-y>1$

Which of the following ordered pairs (x, y) satisfies the system of inequalities above?

A. $(-2,-1)$

B. $(-1,3)$

C. $(1,5)$

D. $(2,-1)$

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43 028867-DC · Medium

$H = 120p + 60$

The Karvonen formula above shows the relationship between Alice's target heart rate H, in beats per minute (bpm), and the intensity level p of different activities. When $p = 0$, Alice has a resting heart rate. When $p = 1$, Alice has her maximum heart rate. It is recommended that p be between 0.5 and 0.85 for Alice when she trains. Which of the following inequalities describes Alice's target training heart rate?

A. $120 \leq H \leq 162$

B. $102 \leq H \leq 120$

C. $60 \leq H \leq 162$

D. $60 \leq H \leq 102$

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44 a19c43f7-f093-4542-805a-9d67d7af0efa · Medium

An event planner is planning a party. It costs the event planner a onetime fee of $\$ 35$ to rent the venue and $\$ 10.25$ per attendee. The event planner has a budget of $\$ 300$. What is the greatest number of attendees possible without exceeding the budget?

Student-produced response: _____

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45 4e92e788-6fae-4f7c-a838-84939a7bdd2f · Medium

$$y < -4x + 4$$

Which point (x, y) is a solution to the given inequality in the *xy*-plane?

A. $(-4, 0)$

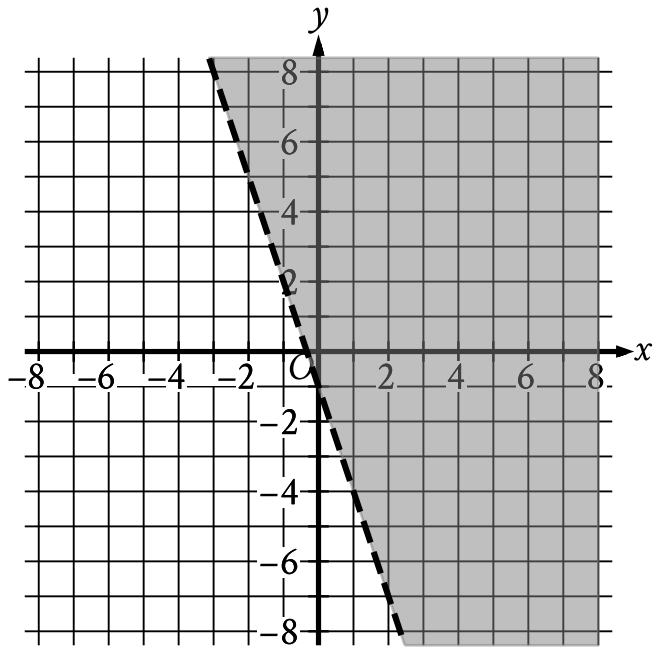
B. $(0, 5)$

C. $(2, 1)$

D. $(2, -1)$

A number x is at most 17 less than 5 times the value of y. If the value of y is 3, what is the greatest possible value of x?

Student-produced response: _____



- The boundary of the inequality is a dashed line.
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2)
    - (0 comma negative 1)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y < -1 + 3x$

B. $y < -1 - 3x$

C. $y > -1 + 3x$

D. $y > -1 - 3x$

48 4f8c6caa-f616-4342-98a4-ffa24734a560 · Medium

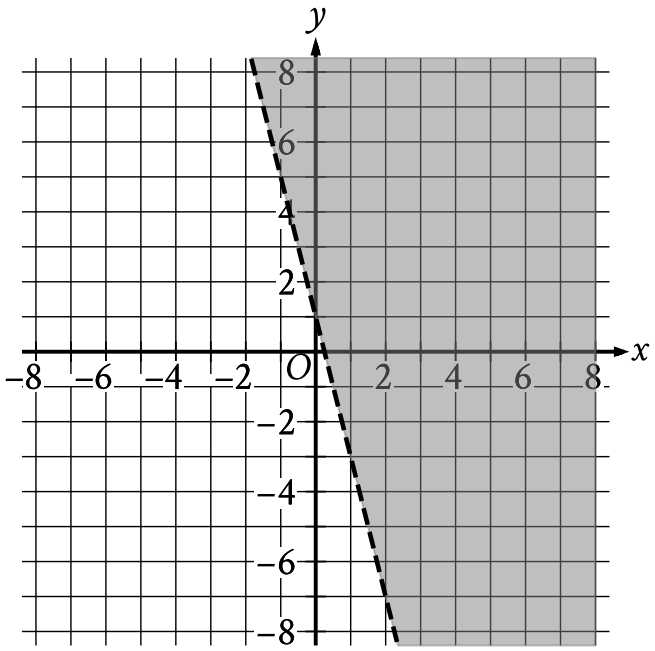
To complete a landscaping project, Adam charges a fee of $\$ 20.00$ for his equipment and $\$ 9.50$ per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of $\$ 17.00$ for her equipment and $\$ 10.00$ per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

A. $5 \leq x \leq 6$

B. $6 \leq x \leq 7$

C. $x < 5$

D. $x > 6$



- The boundary of the inequality is a dashed line.
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 5)
    - (0 comma 1)
    - (1 comma negative 3)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y < 1 + 4x$

B. $y < 1 - 4x$

C. $y > 1 + 4x$

D. $y > 1 - 4x$

50 51deab81-34e8-49ee-b5a8-f1a85ae8f283 · Medium

$$y > 4x + 8$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
2	19
4	30
6	41

B.

x	y
2	8
4	16
6	24

C.

x	y
2	13
4	18
6	23

D.

x	y
2	13
4	21

6	29

51 ea1a5afe-c544-44e2-a9c7-b0921f19d7a7 · Medium

A truck can haul a maximum weight of $5{,}630$ pounds. During one trip, the truck will be used to haul a 190-pound piece of equipment as well as several crates. Some of these crates weigh 25 pounds each and the others weigh 62 pounds each. Which inequality represents the possible combinations of the number of 25-pound crates, x, and the number of 62-pound crates, y, the truck can haul during one trip if only the piece of equipment and the crates are being hauled?

A. $25x + 62y \leq 5{,}440$

B. $25x + 62y \geq 5{,}440$

C. $62x + 25y \leq 5{,}630$

D. $62x + 25y \geq 5{,}630$

52 39ee95df-7fb6-49e1-a615-ef2783d330a7 · Medium

Terry has a goal to collect at least 16 items per week for a donation drive. This week, Terry has collected 4 items. If x represents the additional number of items Terry needs to collect this week to meet the goal, which inequality represents this situation?

A. $4 - x \leq 16$

B. $4 + x \leq 16$

C. $4 - x \geq 16$

D. $4 + x \geq 16$

A model estimates that whales from the genus *Eschrichtius* travel 72 to 77 miles in the ocean each day during their migration. Based on this model, which inequality represents the estimated total number of miles, x, a whale from the genus *Eschrichtius* could travel in 16 days of its migration?

A. $72 + 16 \leq x \leq 77 + 16$

B. $7216 \leq x \leq 7716$

C. $72 \leq 16 + x \leq 77$

D. $72 \leq 16x \leq 77$

54 f945d185-c981-4580-acc0-7976d9988dbb · Medium

$$2x - y > 883$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
440	0
441	-2
442	-4

B.

x	y
440	0
442	-2
441	-4

C.

x	y
442	0
440	-2
441	-4

D.

x	y
442	0
441	-2

440	-4

$$y < x$$

$$x < 22$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A.

x	y
19	18
20	19
21	20

B.

x	y
19	20
20	21
21	22

C.

x	y
23	22
24	23
25	24

D.

x	y
23	24

24	25
25	26

56 1678f18c-082f-4417-97ca-e3e524d586e6 · Medium

A moving truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than $4{,}600$ pounds. What is the maximum number of boxes this truck can tow in a trailer with a weight of 500 pounds if each box weighs 120 pounds?

A. 34

B. 35

C. 38

D. 39

57 e82ae7c1-3cdb-480b-8a7b-5d3fe9c18b84 · Medium

$$y > 14$$

$$4x + y < 18$$

The point $x, 53$ is a solution to the system of inequalities in the xy-plane. Which of the following could be the value of x?

A. -9

B. -5

C. 5

D. 9

58 022222-DC · Medium

A cargo helicopter delivers only 100-pound packages and 120-pound packages. For each delivery trip, the helicopter must carry at least 10 packages, and the total weight of the packages can be at most 1,100 pounds. What is the maximum number of 120-pound packages that the helicopter can carry per trip?

A. 2

B. 4

C. 5

D. 6

---

59 c72f11a1-3ee5-4aee-a4eb-ef3be78817bb · Hard

A number x is at most 2 less than 3 times the value of y. If the value of y is -4, what is the greatest possible value of x?

Student-produced response: _____

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60 176e3a8c-9f87-406f-a76a-01267378721c · Hard

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side lengths of 6 and 12, which inequality represents the possible lengths, x, of the third side of the triangle?

A. $x < 18$

B. $x > 18$

C. $6 < x < 18$

D. $x < 6$ or $x > 18$

---

61 7e2ec609-ff10-4d96-85a3-7a8db042e0c0 · Hard

A salesperson's total earnings consist of a base salary of x dollars per year, plus commission earnings of 11% of the total sales the salesperson makes during the year. This year, the salesperson has a goal for the total earnings to be at least 3 times and at most 4 times the base salary. Which of the following inequalities represents all possible values of total sales s, in dollars, the salesperson can make this year in order to meet that goal?

A. $2x \le s \le 3x$

B. $\frac{2}{0.11}x \le s \le \frac{3}{0.11}x$

C. $3x \le s \le 4x$

D. $\frac{3}{0.11}x \le s \le \frac{4}{0.11}x$

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62 09274-DC · Hard

Adam's school is a 20-minute walk or a 5-minute bus ride away from his house. The bus runs once every 30 minutes, and the number of minutes, w, that Adam waits for the bus varies between 0 and 30. Which of the following inequalities gives the values of w for which it would be faster for Adam to walk to school?

A. $w - 5 < 20$

B. $w - 5 > 20$

C. $w + 5 < 20$

D. $w + 5 > 20$

---

63 f77e335f-77c7-457b-b90f-ba4526c10c28 · Hard

$$y > 13x - 18$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	21
5	47
8	86

B.

x	y
3	26
5	42
8	86

C.

x	y
3	16
5	42
8	81

D.

x	y
3	26
5	52

x	y
8	91

---

64 ceb288c3-af1c-43f5-a67d-ef140ccf15e3 · Hard

A team hosting an event to raise money for new uniforms plans to sell at least 140 tickets before this event and at least 220 tickets during this event to raise a total of at least $\$5{,}820$ from all tickets sold. The price of a ticket during this event is $\$3$ less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?

A. $140x + 3 + 220x \leq 5{,}820$

B. $140x + 3 + 220x \geq 5{,}820$

C. $140x - 3 + 220x \leq 5{,}820$

D. $140x - 3 + 220x \geq 5{,}820$

---

65 4ab12ef2-53a0-47fe-ac2b-2bb21f21eb96 · Hard

$$11x + 14y \leq 115$$

Anthony will spend at most $\$115$ to purchase x small cheese pizzas and y large cheese pizzas for a team dinner. The given inequality represents this situation. Which of the following is the best interpretation of $14y$ in this context?

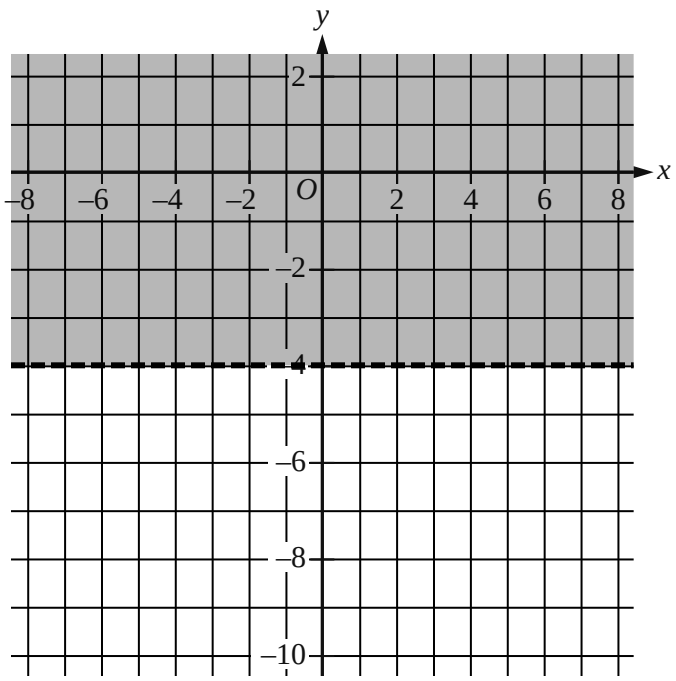
A. The amount, in dollars, Anthony will spend on each large cheese pizza

B. The amount, in dollars, Anthony will spend on each small cheese pizza

C. The total amount, in dollars, Anthony will spend on large cheese pizzas

D. The total amount, in dollars, Anthony will spend on small cheese pizzas

---



- The boundary of the inequality is a dashed line.
  - The line is horizontal.
  - The line passes through the following points:
    - (negative 3 comma negative 4)
    - (0 comma negative 4)
    - (3 comma negative 4)
- The area above the boundary is shaded.

The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r?

A. 15

B. 4

C. -4

D. -15

67 57e960ad-dbb1-486d-9f07-1da462789c42 · Hard

$$y \leq x + 7$$

$$y \geq -2x - 1$$

Which point x, y is a solution to the given system of inequalities in the xy-plane?

A. $-14, 0$

B. $0, -14$

C. $0, 14$

D. $14, 0$

---

68 05385-DC · Hard

A shipping service restricts the dimensions of the boxes it will ship for a certain type of service. The restriction states that for boxes shaped like rectangular prisms, the sum of the perimeter of the base of the box and the height of the box cannot exceed 130 inches. The perimeter of the base is determined using the width and length of the box. If a box has a height of 60 inches and its length is 2.5 times the width, which inequality shows the allowable width x, in inches, of the box?

A. $0 < x \leq 10$

B. $0 < x \leq 11\frac{2}{3}$

C. $0 < x \leq 17\frac{1}{2}$

D. $0 < x \leq 20$

---

69 b56c5e06-9287-4cf4-860c-261d954a284d · Hard

A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is $\$ 14{,}000$, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

A. $\$ 148.15$

B. $\$ 161.53$

C. $\$ 172.84$

D. $\$ 184.94$

---

70 11266-DC · Hard

$$y > 2x - 1$$
$$2x > 5$$

Which of the following consists of the y-coordinates of all the points that satisfy the system of inequalities above?

A. $y > 6$

B. $y > 4$

C. $y > \frac{5}{2}$

D. $y > \frac{3}{2}$

---

71 4761cfd8-02ac-4622-bea3-45afe1718e1e · Hard

$$y < 5x + 6$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	17
5	27
7	37

B.

x	y
3	17
5	35
7	37

C.

x	y
3	25
5	35
7	45

D.

x	y
3	21
5	31

7	41

72 05355-DC · Hard

Ken is working this summer as part of a crew on a farm. He earned $8 per hour for the first 10 hours he worked this week. Because of his performance, his crew leader raised his salary to $10 per hour for the rest of the week. Ken saves 90% of his earnings from each week. What is the least number of hours he must work the rest of the week to save at least $270 for the week?

A. 38

B. 33

C. 22

D. 16

73 03867-DC · Hard

$$I = \frac{V}{R}$$

The formula above is Ohm's law for an electric circuit with current I, in amperes, potential difference V, in volts, and resistance R, in ohms. A circuit has a resistance of 500 ohms, and its potential difference will be generated by n six-volt batteries that produce a total potential difference of $6n$ volts. If the circuit is to have a current of no more than 0.25 ampere, what is the greatest number, n, of six-volt batteries that can be used?

Student-produced response: _____

74 05431-DC · Hard

A local transit company sells a monthly pass for $95 that allows an unlimited number of trips of any length. Tickets for individual trips cost $1.50, $2.50, or $3.50, depending on the length of the trip. What is the minimum number of trips per month for which a monthly pass could cost less than purchasing individual tickets for trips?

Student-produced response: _____

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75 3e613a2d-72f8-4e08-8fce-8abf51849968 · Hard

A small business owner budgets $\$ 2{,}200$ to purchase candles. The owner must purchase a minimum of 200 candles to maintain the discounted pricing. If the owner pays $\$ 4.90$ per candle to purchase small candles and $\$ 11.60$ per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

Student-produced response: _____

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76 40946085-fbf1-4c4a-977f-931fd79ef80c · Hard

$$y < 6x + 2$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	20
5	32
7	44

B.

x	y
3	16
5	36
7	40

C.

x	y
3	16
5	28
7	40

D.

x	y
3	24
5	36

7	48

77 901a9121-4094-4653-b9b7-79d1132f3e1e · Hard

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x. The product of 12 and the fourth odd integer is at most 26 less than the sum of the first and third odd integers. Which inequality represents this situation?

A. $12x + 6 \leq x + x + 4 - 26$

B. $12x + 6 \geq 26 - x + x + 4$

C. $12x + 4 \leq x + x + 3 - 26$

D. $12x + 4 \geq 26 - x + x + 3$